Hamza Mohammed Hassanain

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Cloud & Full-Stack Engineer

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Education

B.Sc. Computer and Information Sciences, *Suez Canal University* 2022 – 2026
AWS Certified Solutions Architect – Associate, *Amazon Web Services* verify on Credly

Experience

Backend & Infrastructure Engineer **Nov 2025 – Present**
Repovive *Part Time – United States, Remote*
A competitive-programming platform pairing scalable automated judging infrastructure and contest tooling with AI-assisted problem authoring.

- Architected the judge infrastructure (`judge1` on **Judge0**) for sandboxed, multi-language auto-grading at contest scale.
- Scaled concurrent execution **10x (50 to 500+ processes)** with an async, containerized job-orchestration pipeline.
- Cut cloud infrastructure cost **95% ($600 to $25/mo)** by migrating serverless workloads to a Docker/Nginx/Redis stack.
- Sandboxed **20+ languages** with resource- and syscall-limited Judge0 Docker images for secure multi-tenant contests.
- Built a real-time collaborative code workspace using **WebSockets** and **Yjs CRDTs** over a Monaco editor.
- Co-built **Build v2**, an AI problem-authoring IDE with a 28-tool agent pipeline guarded by server-side execution locks.
- Tech: TypeScript, Next.js, AWS, Docker, Redis, MongoDB, Nginx, Yjs.

Open Source Contributions

- **GitLab Go SDK (client-go):** typed group integrations – Slack, Jira, MS Teams, Mattermost, Harbor, +3 | **!2692**, **!2691**, **!2675**, **!2670**.
- **GitLab Terraform provider:** new Harbor & Mattermost integration resources + group SAML attribute | **!2881**, **!2909**, **!2852**.
- **GitLab monorepo:** `Json.safe_parse` hardening and Vue lint compliance | **!221163**, **!221158**.
- **LLVM / Clang:** constexpr X86 `cvtpd2ps` eval and AArch64/ARM CLS → `ISD::CTLS` lowering | **#169980**, **#178885**, **#178430**.
- **Apple Swift NIO:** idempotent `NIOFileSystem` directory creation | **#3410**.

Projects

Competitive-Programming Judge — AWS Reference Architecture **github.com/HamzaHassanain/CP-Judge-AWS**
- Designed an AWS reference architecture: Judge0 `isolate` on ECS-on-EC2, Fargate orchestrator, Spot ASG on queue depth.
- Automated post-round cheating detection with Step Functions, AWS Batch, Lambda/Bedrock, and Athena.

Autonomous AI Tutor & Curriculum Engine, Graduation Project **github.com/sigma-loop**
- Led backend & DevOps for an 8-person team building an on-demand AI course, lesson, and challenge generator.
- Graded code in a Judge0 sandbox and math proofs via a confidence-scored LLM; owned all CI/CD and AWS deployments.

Polyman CLI **npmjs.com/package/polyman-cli | github.com/HamzaHassanain/polyman**
- Built a TypeScript CLI automating the Codeforces Polygon workflow: test generation, validation, and direct API sync.

Systems in C++ — HAllocator & cppress **HAllocator | cppress**
- Built a Red-Black-Tree best-fit memory allocator (**O(log n)**) and an epoll-driven C++ web framework, both from scratch.

Technical Writing

- Authored ***"What Every Programmer Should Know About Memory"***, a series on CPU caches, virtual memory, and optimization.
- Published ***"I Built a Backend Web Framework from Scratch in C++"***, covering non-blocking I/O, HTTP parsing, and thread pools.

Achievements

ICPC ECPC & ACPC — Competitive Programming **icpc.global/YC7Z02WQCPIB**
***17th** of **500** teams at ECPC (2024); **two-time ACPC regional** qualifier (**2024, 2025)**, top **250** in the Africa & Arab region.*

Skills

Cloud & DevOps	AWS (Solutions Architect – Associate), Docker, Nginx, Redis, Terraform, CI/CD (GitHub Actions), DigitalOcean, Linux
Backend & Full-Stack	TypeScript, Node.js, Express.js, Next.js, React, MongoDB, PostgreSQL, REST APIs, Microservices, WebSockets, CRDTs (Yjs)
Languages & Systems	C++, Go, Rust, Python, SQL; System Design, Distributed Systems, Message Queues